

Mohammed Eid Abdelmeguid

📍 Alexandria, Egypt

✉ MohammedEidAbdelmeguid@gmail.com

☎ +20 12 10649832

🐙 [GitHub](#)

🌐 [LinkedIn](#)

🌐 [Portfolio](#)

📖 [Hugging Face](#)

🏆 [Kaggle](#)

Summary

AI/ML Engineer in the final stage of a 9-month AI & Data Science diploma, with hands-on experience designing and shipping full-stack AI systems end to end: a multi-agent RAG platform with a custom recommendation engine, and an Arabic NLP knowledge-graph and retrieval pipeline. Comfortable across the ML lifecycle — data ingestion, feature engineering, model evaluation, and deployment design — backed by a software engineering foundation in Python, Django, REST APIs, and PostgreSQL that translates directly into shipping working AI systems rather than notebooks. Seeking an AI Engineer / Machine Learning Engineer / Data Scientist role to apply production-oriented Machine Learning, NLP, and LLM/RAG system design.

Skills

Programming Languages: Python, SQL, C

AI & Machine Learning: Scikit-learn, XGBoost, Supervised/Unsupervised Learning, Feature Engineering, Model Evaluation & Benchmarking

Deep Learning: TensorFlow, Keras, Neural Networks, Computer Vision

NLP: CAMEL Tools (Arabic NLP), NLTK, Named Entity Recognition, Topic Modeling (BERTopic), Arabic Text Normalization

LLM & RAG: Retrieval-Augmented Generation, Vector Databases (pgvector), Embeddings (Sentence-Transformers), Multi-Agent Systems, Prompt Engineering, Structured Outputs / JSON Schema, Knowledge Graphs (NetworkX, Wikidata Entity Linking), LLM Platforms & Serving (Groq API, Ollama)

Data Science: Pandas, NumPy, Matplotlib, Plotly, Exploratory Data Analysis, Statistical Evaluation, Time-Series Analysis

Data Engineering & Backend: Django, FastAPI, Flask, REST APIs, Celery, SQLAlchemy, Web Scraping (Scrapy), ETL Pipelines

Databases: PostgreSQL, pgvector, Redis, SQL

MLOps & Evaluation: Offline Evaluation Harnesses (NDCG@10, MAP, Recall@k, F1), Ablation Testing, Schema-Validated Pipelines (Pydantic)

Tools & Version Control: Git, GitHub, GitHub Actions, Jupyter Notebook, Linux, uv

Projects

AI Compass — Multi-Agent AI News & Recommendation Platform

May 2026 – Present

- Led a cross-functional team of five members as Team Lead and primary technical contributor, owning the system architecture, AI pipeline, backend/database design, and deployment plan for a multi-service AI news platform (Python/SQLAlchemy pipeline, Django 5.2 backend, Next.js 16 frontend) that ingests content from 7 sources (arXiv, GitHub, Hugging Face, YouTube, RSS, blogs).
- Designed a 5-agent enrichment and RAG pipeline (local Sentence-Transformer embeddings, PostgreSQL/pgvector, Groq-served LLMs) powering semantic search and a streaming RAG chat assistant.
- Built a deterministic two-stage recommendation engine — pgvector-similarity and rule-based candidate generation, multi-factor weighted scoring, and MMR diversification — validated with a custom offline NDCG@10/MAP evaluation harness.
- Owned the async infrastructure (Celery + Redis across 3 dedicated queues with a 6-job scheduler) and designed a production-ready, containerized deployment architecture — a multi-service Docker Compose topology with Caddy-routed TLS across three services on one domain — with a documented \$0/month path to Oracle Cloud or Render.

Palestinian Cultural Knowledge Platform — Arabic NLP, RAG & Knowledge Graph

Jun – Aug 2026

- Collaborative NLP research project (2-person team) turning a quality-scored, deduplicated Arabic/English corpus on Palestinian history and culture, collected from 5 sources (Wikipedia, Semantic Scholar, GDELT, WAFA), into structured, queryable knowledge; primary technical contributor.
- Designed the ingestion pipeline (schema-validated records, composite quality scoring, persistent MinHash/LSH deduplication) and led Arabic NLP (CAMEL Tools NER, F1 0.47) and RAG (embeddings → pgvector → locally-served LLM generation, Recall@5 0.93 over 1,282 indexed passages).
- Built the knowledge-graph pipeline (entity canonicalization, Wikidata entity linking, LLM-based relation extraction) producing a 13,063-entity, 187-edge graph, plus bias and topic analysis (WEAT, BERTopic) exposed through a FastAPI REST API and a Streamlit dashboard.
- Published a bilingual (Arabic/English) cultural knowledge dataset — 882 documents, ~890K words — publicly on [Hugging Face Datasets](#) .

AynVQA-MSA — Arabic Spoken Visual Question Answering

Jul – Aug 2026

- Built an end-to-end multimodal AI pipeline (audio, vision & language) for the **ImageEval 2026 / ArabicNLP** shared task (Arabic Spoken Visual Question Answering): Whisper ASR → schema-constrained LLM parsing of spoken questions/options → conditional repair escalation → vision-language-model multiple-choice selection.
- Ran systematic vision-language-model and prompting experiments (model comparison, chain-of-thought and few-shot ablations), reaching 79.00% accuracy (395/500) on the official development split — nearly double the organizers' baseline (39.8%).

Fitness Activity Tracker — ML-Powered Exercise Classification

Mar – Apr 2026

- 3-person team project (Digilians AI & Data Science Diploma) classifying barbell exercises and counting repetitions from wrist-worn accelerometer/gyroscope sensor data.
- Owned data preprocessing, feature engineering (Chauvenet outlier removal, PCA, FFT-based frequency-domain features, k-means clustering), exploratory analysis, and the model evaluation pipeline — compared 5 classifiers via grid search, with Random Forest reaching the best held-out test accuracy (~99%).
- **Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib

Education

Faculty of Engineering, Alexandria University	Sep 2019 – Jun 2024
BS in Communication and Electronics Engineering	
- CGPA: 3.13 / 4.0 - Graduation Project: Advanced Driver Assistance Systems (ADAS) — Grade A+, ranked 5th of ~60 graduation projects (Value competition). Team Lead; owned the software and computer-vision modules (lane keeping, automatic emergency braking, sign detection). Mentored by Valeo, sponsored by ITIDA’s Egypt Makes Electronics (EME) program.	

Certifications & Training

AI & Data Science Diploma — Digilians (MCIT / Egyptian Military Academy scholarship) (expected) <i>9-month intensive program: Machine Learning, Deep Learning, Computer Vision, NLP, Data Engineering, MLOps, Generative AI, LLMs</i>	Dec 2025 – Sep 2026
IBM AI Engineering Professional Certificate — Coursera <i>Deep Learning, Neural Networks, Computer Vision, NLP, Model Deployment with PyTorch & Keras</i>	Completed
Machine Learning Specialization — Andrew Ng / Coursera <i>Supervised & Unsupervised Learning, Neural Networks, Practical ML Best Practices</i>	Completed
Data Engineering Professional Program — DEPI	Apr 2024 – Oct 2024
AWS Certified Machine Learning Engineer – Associate (MLA-C01)	In Progress (Self-Study)

Work Experience

Freelance Backend Developer — StockGo (Qatar)	Nov 2024 – Apr 2025
Built and deployed a Django REST API and business-analytics dashboard for a live client product used for inventory tracking (thestockgo.com).	
Backend Developer Intern — Venture Appital	Oct 2024 – Feb 2025
Built modular RESTful microservices in Flask with full unit-test coverage.	
Full Stack Developer Intern — NfertiX	Mar 2025 – Apr 2025
Built an internal Django/JavaScript web application to streamline internal operations.	

Languages

Arabic (Native)		English (Professional)
-----------------	--	------------------------